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Overview of Artificial Intelligence

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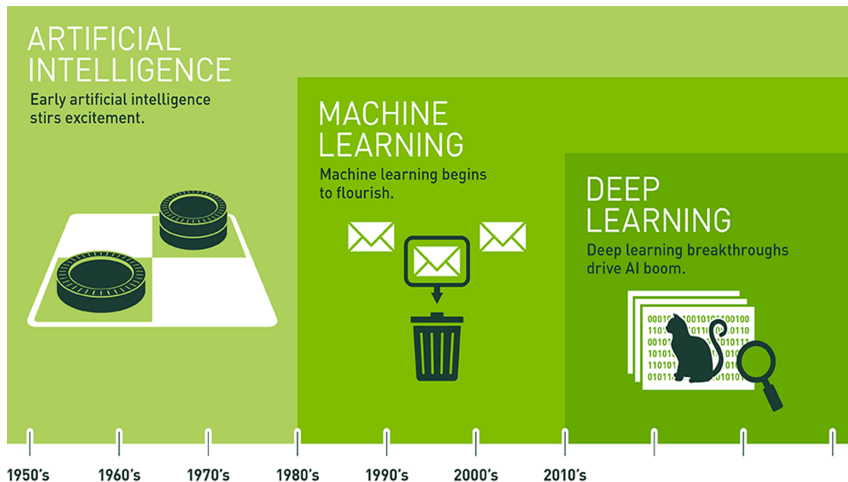


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What is AI?

- Artificial Intelligence (AI) coined in 1956 by John McCarthy.
- An AI is a machine that is aware of its environment and performs **intelligent actions in a goal oriented matter**.
- An AI is a machine that possesses **cognitive capabilities** like learning, planning, reasoning, and problem solving.
- Sub-fields of AI:
 - robotics, speech and language processing, automatic reasoning, autonomous agents, computer vision
 - *Machine Learning, Deep Learning*
- Successes of Deep Learning in computer vision and **language processing** have created the AI hype of the last years.
- AI is a key technology of the coming years.

Taxonomy of AI



Source: NVIDIA

Narrow vs. General AI

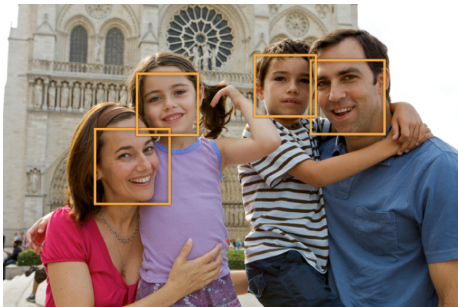
Strong AI / General AI / Artificial General Intelligence (AGI):

a machine that can acquire **intellectual capabilities similar or superior to a human**; possesses *world knowledge*; does not only react, but can act with a will of its own, intelligently, and flexibly.

“Weak” AI / Narrow AI: systems developed for solving **specific tasks**.

Status Quo: all AI solutions that have been developed so far are not general, but application-focused;
projected time line for AGI: +20--40 *years from now*.

Where Do We Meet AI In Everyday Life?



Sources: (1) developer.apple.com | (3) Vagner Vidal/INS News Agency Ltd. | (4) Wall Street Journal | (5) Uwe Rattay/ADAC

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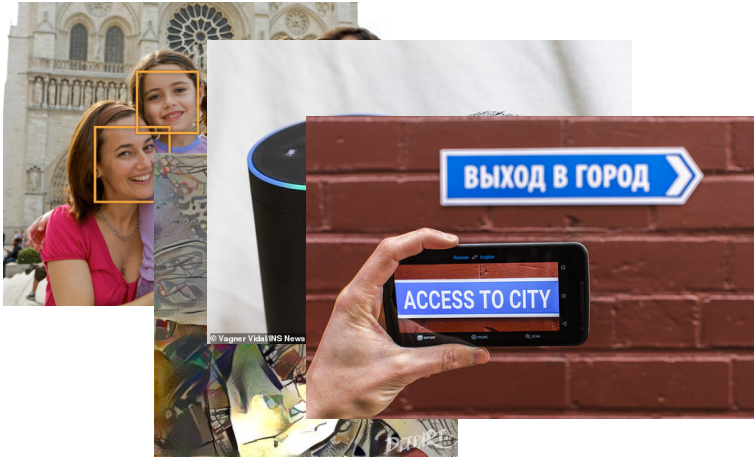
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Where Will We See More AI In The Near Future?



Sources: (1) flatworld solutions | (2) Healthcare Business Today | (3) Haus der Barmherzigkeit/APA-Fotoservice/Pauty |

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Examples in Medicine

- LICA - Using AI to extract relevant information from **medical documentation** and **medical guidelines** for home nursing care.
- Metabolomics Lunge - detection of **pulmonary hypertension** via analysis of chemical biomarkers.
- Treetop Medical - Using AI to extract **treatment processes** from medical documentation.

Machine Learning == Learning From Data



0 4 1 9 2 1 3 1 4 3
5 3 6 1 7 2 8 6 9 4
0 9 1 1 2 4 3 2 1 3
8 6 9 0 5 6 0 7 6 1
8 7 9 3 9 8 5 9 3 3
0 7 4 9 8 0 9 4 1 4
4 6 0 4 5 6 1 0 0 2
7 1 6 3 0 2 1 1 7 9
0 2 6 7 8 3 9 0 4 6
7 4 6 8 0 7 8 3 1 5

Source: pixabay